Hyungryul Baik (Course Coordinator)

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Course Code: AIM.60003

Course Title: AI-Based Discovery in Mathematics
(인공지능 기반 수학 발견)

Day(s), Time: TuTh, 10:30-12:30

Place: Creative Learning Bldg. (E11, 창의학습관), Room 210

Teaching Assistant: Hyeong Jun Jang (kael21ce@kaist.ac.kr)

Course Description: This course explores how modern artificial intelligence interacts with mathematical discovery, and builds the mathematical foundations needed to understand, use, and advance AI. The course is organized as a series of block lectures given by three instructors:

- Prof. Eunjin Jung (University of San Francisco) — lectures delivered via Zoom
- Prof. Sang-hyun Kim (KIAS)
- Prof. Geon Ho Choe (KAIST)

Details of each instructor's lectures are given on the following pages.

Prerequisite(s): N/A
Language: English

Grade Distribution:

Attendance	80%
Project	20%

- **Attendance (80 points):** 5 points are deducted from 80 for each absence. Deductions apply regardless of the reason for the absence. One or two absences will not affect your final grade.
- **Project (20 points):** The AI-assisted mathematical exploration project guided by Prof. Sang-hyun Kim. After his first lecture (Oct 13), each student chooses a topic and carries out the exploration, then presents and discusses the results in his last lecture (Nov 17). See the lecture plan of Prof. Kim for details.

Course Policies:

- **Note on class hours:** the official course registration system lists the class time as 10:30–12:00, but each lecture will actually run until **12:30**. Please plan your schedule accordingly.
- All announcements will appear on KLMS.
- Prof. Eunjin Jung's lectures will be given via Zoom (the link will be announced).

Lecture Schedule (Fall 2026):

Tuesday	**Instructor**	**Thursday**	**Instructor**
Sep 1	Eunjin Jung	Sep 3	Geon Ho Choe
Sep 8	Eunjin Jung	Sep 10	Geon Ho Choe
Sep 15	Eunjin Jung	Sep 17	Geon Ho Choe
Sep 22	Eunjin Jung	Sep 24	*No class (Chuseok)*
Sep 29	Eunjin Jung	Oct 1	Geon Ho Choe
Oct 6	Eunjin Jung	Oct 8	Geon Ho Choe
Oct 13	Sang-hyun Kim	Oct 15	Geon Ho Choe
Oct 20	*No class (midterm week)*	Oct 22	*No class (midterm week)*
Oct 27	Sang-hyun Kim	Oct 29	Geon Ho Choe
Nov 3	Sang-hyun Kim	Nov 5	Geon Ho Choe
Nov 10	Special lecture (speaker TBA)	Nov 12	Geon Ho Choe
Nov 17	Sang-hyun Kim	Nov 19	Geon Ho Choe

The last lecture is on November 19 (the course ends earlier than the official end of the semester).

Lectures by Prof. Eunjin Jung (Tuesdays, via Zoom):

Six lectures (Sep 1 – Oct 6) covering the following modules in order. (Logic-based models are not covered here; first-order logic is treated in depth in Prof. Sang-hyun Kim's lectures.)

Module 0: Using Google Colab

- Introduction to Jupyter Notebook and Google Colab (no prior experience assumed)

Module 1: State-Based Models

- Basic tree search (BFS, DFS)
- Uniform cost search (Dijkstra's algorithm)
- Heuristic search (A*)
- Adversarial search

Module 2: Machine Learning

- K-means
- PCA
- Linear regression
- Logistic regression
- CART

Module 3: Variable-Based Models

- Bayesian networks
- Markov models

Lectures by Prof. Geon Ho Choe (Thursdays):

An intensive, mathematically rigorous exploration of the continuous, geometric, and algebraic structures that power modern complex systems and computational models. Theoretical development is paired with Python simulations.

Date	**Topics**	**Why it matters for AI**
Sep 3	Information theory: Shannon's information content; data compression (Huffman, Ziv–Lempel)	Entropy is a fundamental tool in training language models
Sep 10	Linear algebra I: eigenvalues, SVD, tensor products of matrices	Tensors in deep learning; SVD underlies PCA, low-rank approximation, model compression
Sep 17	Linear algebra II: exponential and logarithm of matrices	Continuous-time models; Lie-theoretic foundations for spatial transformations
Oct 1	Ordinary differential equations: vector fields, autonomous linear systems, nonlinear dynamics	Gradient descent as a continuous flow along a vector field
Oct 8	Partial differential equations: heat equation	Diffusion of energy over time is closely related to diffusion models
Oct 15	Maxwell's equations: vector calculus, electromagnetism	Divergence, curl, and gradient fields in scientific ML (e.g., PINNs)
Oct 29	Quaternions: 3D rotation by quaternion multiplication	Robust, efficient 3D rotations in robotics, spatial AI, and 3D vision
Nov 5	Numerical analysis: numerical linear algebra, numerical integration of ODEs	Runge–Kutta-type methods for continuous-time AI models
Nov 12	Probability and stochastic processes: measure-theoretic probability, Markov chains, Brownian motion, SDEs	Stochastic framework for modeling noise and uncertainty in AI
Nov 19	Non-Euclidean and hyperbolic geometry: Poincaré disk and upper half-plane models, hyperbolic metric	Exponential volume growth suits embeddings of hierarchical data

Lectures by Prof. Sang-hyun Kim (four lectures):

AI-based discovery in mathematics: lecture plan.

Lecture 1 (Oct 13). Artificial intelligence in mathematical research: trends and case studies

- Recent research trends (2021–2026): machine-learning-based conjecture generation, FunSearch, AlphaEvolve, AlphaProof, etc.
- The structure of search methods combining language models with evaluation functions
- Case studies: the Aletheia agent, the Erdős experiments, the refutation of an Erdős unit-distance conjecture
- Project guidance: each student selects a topic for AI-assisted mathematical exploration and submits a plan (including an autonomous verification scheme)

Lecture 2 (Oct 27). First-order logic I: foundations

- Syntax and semantics of first-order logic; theories and models
- Completeness theorem, compactness theorem, Löwenheim–Skolem theorems
- Decision problems and incompleteness: Gödel's incompleteness theorems; examples of decidable theories (Presburger arithmetic, real closed fields)

Lecture 3 (Nov 3). First-order logic II: model theory and groups

- Quantifier elimination; ultraproducts and the theorem of Łoś; applications to asymptotic cones
- First-order theories of groups: Tarski's problem on free groups, first-order rigidity of lattices
- Decision problems in group theory: undecidability of the word problem and the isomorphism problem (Novikov–Boone, Adian–Rabin)

Lecture 4 (Nov 17). Verification, autonomous research, and sharing of results

- Overview of formal proofs and machine verification (Lean, etc.)
- Verifying mathematics generated by language models: cross-verification with multiple models, computational certificates
- Methodology of autonomous research: problem selection, research logs and handover documents, verification of results
- Project presentations and discussion; changes in mathematical research and education

Project: After Lecture 1, each student chooses a topic and carries out an AI-assisted mathematical exploration, then presents and discusses the results in Lecture 4.